

Irina Nicolae

Lead machine learning researcher

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Selected work experience

2019–now **Researcher, machine learning & security, Bosch Research, Stuttgart, Germany**

LLM trust Project lead—trust for LLM applications (ongoing)

- Automated the evaluation of LLM applications by developing methods, metrics and tools.
- Established “Risks of GenAI” as strategic topic for Bosch Research.

Fuzz testing Automated software testing using ML

- Increased fault discovery in CAN protocol fuzzing by 20x by generating test data from a transformer.
- Identified fundamental limitations in neural program smoothing for fuzzing.

Planar robots Hybrid ML control for magnetic levitation

- Improved levitation performance in a PID controller by training a model on an inverse problem.

Power tools Embedded ML for electric screwdrivers

- Implemented an automated screw stop feature by training an ML model on time series.

2017–2019 **Researcher, adversarial ML, IBM Research, Dublin, Ireland**

- Created and led the IBM Adversarial Robustness Toolbox, one of the top libraries in the field.
- Developed new defenses against adversarial attacks and model robustness metrics.
- Won Darpa grant, IBM Outstanding Technical Accomplishment Award and EU Horizon projects.

Education

2013–2016 **Ph.D. in machine learning, Jean Monnet University, France**

- Developed new metric learning algorithms for classification on tabular data and time series, with theoretical guarantees on their performance.

2011–2013 **MSc in computer science, Ensimag, Grenoble INP, France, Major in Information Systems**

2007–2011 **BSc in computer science, Politehnica University of Bucharest, Romania**

Skills

ML paradigms Supervised, unsupervised and self-supervised learning

ML tools TensorFlow, Keras, PyTorch, ScikitLearn, Numpy, Pandas

Programming Python, C

Languages English, French, Romanian

Deep learning CNNs, RNNs, autoencoders, GANs, transformers

LLM tools LangChain, LangGraph, HuggingFace, OpenAI, LiteLLM

DevOps Docker, GitHub Actions, GitLab

Interpersonal Driven, pragmatic, honest, team player

Other accomplishments

Publications **Academic** ICML, ICLR, NeurIPS, ESEC/FSE, CCS, AAI, ECML/PKDD.

Industrial BlackHat, RSA, Dagstuhl and over 15 patents.

Reviewer ICLR 2021–, NeurIPS 2016–, ICML 2016–, AAI 2020–2022, IJCAI 2020–2024.

Editor <https://medium.com/security-garten>: security, privacy and safety blog.

Organizer Nemesis Workshop at ECML/PKDD 2018, IDA International Symposium 2015.

Mentoring N. Niayesh, Z. Zhuang, K. Sheng, J. Messner, A. Soutiff, V. Zantedeschi.